Dependently Reconstructible Distinguishability (DRD): A Functorial Theory of Information Carriers, Graded Spectral Capacities, and Causal Integration on Compact S^3 Geometry

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Abstract

Existing quantitative theories of information are measure theories over presupposed alphabets: Shannon's theory presupposes a symbol set, algorithmic complexity a machine model, quantum information a Hilbert space. This paper develops a definition-first framework—*dependently reconstructible distinguishability* (DRD)—for the layer beneath those theories: what a distinction is, what carries it, which operations preserve it, which destroy it and at what price, and how much of it a physical domain can carry and causally integrate.

The framework is organized in three strictly separated layers. (I) A *mathematical layer*: the correct object is not one flat functor but a fibrewise profile over a category of structures, $\mathbf{Car}_*^{\mathrm{fib}} = \int_{\mathfrak{s}\in\mathbf{Str}} \mathbf{Car}_{\mathfrak{s}}$, with $\mathsf{D}^\sharp_{\mathfrak{s}} : \mathbf{Car}_{\mathfrak{s}} \to \mathbf{Prof}_{\mathfrak{s}}$. Homology, cohomology, characteristic classes, spin data and spectra are preserved only inside the structural fibre whose morphisms preserve the relevant structure. (II) An *erasure and capacity layer*: non-invertible change is measured by a cone-defect package $\mathbb{E}(f)$, sector erasure has the safe Landauer component $k_B T \ln N$ plus platform-dependent geometric cost, and the graded scalar, TT and Dirac capacities on round S^3_R are counted with explicit dimension and cutoff conventions. Heat-trace capacity $\mathcal{N}_P(s,\tau) = \mathrm{Tr}(e^{-sP_\tau})$ replaces discontinuous sharp counts when a reconstruction flow varies the metric. (III) A *physical-selection layer*: causal integration is exact for shell-complete homogeneous spectral projectors, while the only central remaining physical node is the gravitational selector. It is not a ratio between unlike quantities: the raw spectral count N_P is a mode number, while the selected physical object is $\log_2 \dim \mathcal{H}_{\mathrm{code}}$ in bits. The selector is therefore defined only after an operational finite-alphabet or energy/occupation cutoff has converted accessible modes into accessible code bits. It then chooses the area-compatible, gauge-invariant, backreaction-stable code subspace from the volume-scaling kinematic alphabet. The framework does not replace Shannon, Holevo, or Landauer; it supplies the pre-statistical object—the physically realizable alphabet of distinguishable states—that those theories presuppose.

Keywords: theory of information; distinguishability; carriers; functorial invariants; suspension; S^3; Hopf holonomy; spectral capacity; causal integration; Landauer principle; covariant entropy bounds; boundary admissibility.

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1 Introduction

1.1 The problem: information named before it is defined

The word "information" now carries load in thermodynamics, quantum theory, cosmology, and the foundations of computation. Yet the standard quantitative theories are measure theories over *presupposed* alphabets. Shannon's theory [1] quantifies the statistics of a source once a set of distinguishable symbols is given; Kolmogorov complexity [2] quantifies description length once a machine model and a symbol set are given; quantum information theory quantifies state distinguishability once a Hilbert space is given, with the Holevo bound [3] governing accessible classical information. None of them answers the prior questions: what is a distinction, what does it take for a distinction to be physically realized, what operations preserve it, what operations destroy it and at what price, and how much distinguishability a given physical domain can carry at a given resolution—and causally integrate in a given time.

Foundational slogans have tried to close the gap from the other side. Wheeler's "it from bit" [10] makes the bit primitive and matter derivative; Landauer's "information is physical" [5] makes the carrier primitive. Both intuitions are correct in part, and both are incomplete without a formal object that says what a "bit" is before any physics, and what physics must add before that object becomes operational.

This paper supplies such an object and develops it as a research framework with an explicit claim-status discipline. In its strict form, DRD is best understood not as "a theory of information in

general" but as a **theory of invariants of reconstructible distinguishability on categories of carriers**. Information is not identified with statistics, with entropy, or with a quantum state; it is defined as *structure-preserved distinguishability*, which acquires physical realization only after metric, causal order, spin structure, a resolution scale, and a class of admissible morphisms are added. This contradicts neither Shannon, nor Holevo, nor Landauer; it places them on different floors of one tower.

The paper proceeds in three layers:

(I) a **mathematical layer** (Secs. 2–4): the fibred carrier category $\mathbf{Car}_*^{\mathrm{fib}}$, the fibrewise profiles $\mathsf{D}_{\mathfrak{s}}^{\sharp}$, and the conservation, attributed-erasure, and transport theorems;

(II) an **operational layer** (Secs. 5–8): two explicit postulates, exact graded capacities for scalar, tensor, and spinor alphabets on S^3_R, and the causal-integration formula $\mathrm{PIR}(T)$;

(III) an **interpretive and programmatic layer** (Secs. 9–13), explicitly labelled: erasure-cost hypotheses Q_k, carrier-resolved quantum resources, the volume–area problem, related definition-first programmes, and the ontological reading.

The companion papers of this programme supply geometric infrastructure: the suspension-chain analysis of S^3 as minimal spin/Hopf carrier [39], the Ricci-type reconstruction flow with its no-go theorems and Planck-barrier invariant region [40], and the compact causal accessibility structure of light on S^3_R [41]. The present paper is the information-theoretic keystone of that line: it states what the other papers' carriers carry.

1.2 Claim-status convention

Label	**Meaning**
Theorem / Proposition / Lemma	Proved under stated mathematical assumptions.
Physical anchor	Established physics cited for contact, not re-derived here.
Postulate	An operational assumption of the physical layer, explicitly adjustable.
Hypothesis	A proposed law-like statement with specified test platforms; not a theorem.
Ansatz	A proposed quantitative form, not derived from dynamics.
Interpretation	Philosophy-of-physics reading; proves no physics.

1.3 What this paper does not claim

Boundary of the paper

This paper does not replace or amend Shannon's theory, Kolmogorov complexity, the Holevo bound, or Landauer's principle; it supplies the pre-statistical object those theories presuppose. It does not derive physical law from topology; does not derive the value of ℓ_P; does not prove that the observed universe is globally S^3; does not derive gravitational entropy bounds; does not resolve the volume–area tension (it states it as the central open problem); and does not assert any ontological statement as a theorem. Every formula in the operational layer is conditional on the stated postulates. Statements of the form "information is neither created nor destroyed" are made only in the precise form: *within a fixed category of invertible structure-preserving morphisms, the corresponding invariant package is neither created nor erased.* The remaining physical problem is explicitly localized in the selector Σ_{grav}, not hidden inside topological language.

Theorem 1.1 (The DRD closure map)**.** *Under the structural and operational assumptions stated in this paper, the usable information capacity of a sector is not a primitive number but the output of four separated maps:*

$$\mathsf{D}^{\sharp}_{q,\Lambda}(\mathcal{X}) \;\rightsquigarrow\; n^{(P)}_{\mathrm{kin}} \xrightarrow{\mathrm{PIR}_P(T)} n^{(P)}_{\mathrm{causal}} \xrightarrow{b^{(P)}_{\mathrm{mode}}} B^{(P)}_{\mathrm{causal}} \xrightarrow{\Sigma^{(P)}_{\mathrm{grav}}} B^{(P)}_{\mathrm{phys}}.$$

The first object is the fibrewise distinguishability profile; $n^{(P)}_{\mathrm{kin}}$ *and* $n^{(P)}_{\mathrm{causal}}$ *are mode numbers;* $B^{(P)}_{\mathrm{causal}}$ *and* $B^{(P)}_{\mathrm{phys}}$ *are bit counts after a finite alphabet, energy cutoff or code-space cutoff has been specified. The mathematical and spectral parts of the map are computable inside their declared fibres. The only irreducible physical selector left open is* $\Sigma^{(P)}_{\mathrm{grav}}$*, the map from accessible code bits to gauge-invariant, gravitationally dressed, entropy-admissible, backreaction-stable code bits.*

Remark 1.2 (What the closure theorem forbids). Equation (1.1) is a bookkeeping discipline. It forbids three common mistakes: treating a vector-space dimension as a bit count, treating a boundary entropy bound as a replacement of the bulk carrier, and treating a suspension isomorphism in reduced homology as an isomorphism of the entire unstable profile.

2 The Full Distinguishability Profile $\mathsf{D}^{\sharp}$

2.1 Elementary distinction

Definition 2.1 (Elementary distinction)**.** An *elementary distinction* is the two-point space $S^0 = \{-1, +1\}$ with its discrete topology. No metric, phase, direction, or causal structure is assumed at this level. The two points are not two objects located somewhere; they are the two poles of a single relational structure, *this / not-this*. Neither point carries the structure alone: removing either yields a one-point space, which is no longer a distinction.

2.2 Why a single rank profile is not enough

A first-pass invariant is the rank profile $\mathsf{D}_k(X) = \operatorname{rank} \widetilde{H}_k(X;\mathbb{Z})$: D_0 counts independent elementary distinctions, D_1 cyclic (phase-type) distinctions, D_2 enclosure-type distinctions, and so on; for spheres $\mathsf{D}_k(S^n) = \delta_{k,n}$. The rank profile already supports invariance and transport theorems, but it conflates levels of structure: it cannot see torsion, ring structure, unstable homotopy data, bundle topology, or resolution-dependent spectral content. A publishable theory must therefore carry a *two-level functorial package*, not one number.

Definition 2.2 (Topological profile)**.** For a pointed space X of the homotopy type of a finite CW complex,

$$\mathsf{D}^{\sharp}_{\mathrm{top}}(X) \;:=\; \Big(\{\widetilde{H}_k(X;\mathbb{Z})\}_{k\geq 0},\; H^*(X;\mathbb{Z}) \text{ with } \smile,\; \{\pi_k(X)\}_{k\geq 1} \text{ with Whitehead products}\Big), \quad (1)$$

where $\widetilde{H}_k(X;\mathbb{Z}) \cong \mathbb{Z}^{\beta_k} \oplus \mathrm{Tor}_k$ retains torsion. The rank shadow is $\mathsf{D}_k(X) = \beta_k$.

Definition 2.3 (Structured carrier and operational profile)**.** A *structured carrier* is a tuple

$$\mathcal{X} = (X, g, \mathfrak{s}, \mathcal{E}, \varepsilon), \quad (2)$$

where g is a Riemannian metric, $\mathfrak{s}$ a spin structure (when required), $\mathcal{E}$ a finite set of distinguished vector bundles/fields, and $\varepsilon > 0$ a resolution scale. Its *operational profile* is

$$\mathsf{D}^{\sharp}_{\mathrm{op}}(\mathcal{X}) \;:=\; \Big(\mathsf{D}^{\sharp}_{\mathrm{top}}(X),\; \mathcal{C}(\mathcal{X}),\; \mathrm{Spec}(\mathcal{X})\Big), \quad (3)$$

where $\mathcal{C}(\mathcal{X})$ is the package of characteristic classes $(w_i, c_i, p_i, e, \hat{A})$ of the bundles in $\mathcal{E}$, and $\mathrm{Spec}(\mathcal{X}) = (\mathrm{Spec}_0, \mathrm{Spec}_{TT}, \mathrm{Spec}(D_{\mathfrak{s}}))$ collects the spectra (with multiplicities) of the scalar Laplacian, the rough (Bochner) Laplacian on transverse-traceless tensors, and the Dirac operator, truncated at the resolution ε where an operational alphabet is required.

2.3 Functoriality

Theorem 2.4 (Functoriality of the topological profile)**.** *A continuous map $f : X \to Y$ induces $f_* : \widetilde{H}_k(X;\mathbb{Z}) \to \widetilde{H}_k(Y;\mathbb{Z})$ (covariant), $f^* : H^*(Y;\mathbb{Z}) \to H^*(X;\mathbb{Z})$ (contravariant) with $f^*(\alpha \smile \beta) = f^*\alpha \smile f^*\beta$, and $\pi_k(f) : \pi_k(X) \to \pi_k(Y)$ compatible with Whitehead products. Characteristic classes are natural under pullback: $x(f^*E) = f^*x(E)$. Hence $\mathsf{D}^\sharp_{\mathrm{top}}$, and the class package $\mathcal{C}$, are functorial with respect to (admissible) continuous maps.*

Proof. Standard algebraic topology and the theory of characteristic classes [11, 12]. □

Proposition 2.5 (Spectral invariance under (spin-)isometry)**.** *If $f : (X, g) \to (Y, h)$ is an isometry, the scalar and TT-tensor Laplace-type operators are unitarily conjugate, so* Spec_0 *and* Spec_{TT} *coincide with multiplicities; if f is in addition a spin-isometry,* $\mathrm{Spec}(D_{\mathfrak{s}})$ *coincides as well. Thus* Spec *is functorial on the subcategory of (spin-)isometric morphisms, not on all continuous maps.*

Proof. Pullback by an isometry intertwines the Laplace-type operators; a spin-isometry lifts to a unitary on spinor fields intertwining the Dirac operators [13]. □

Remark 2.6 (The point of the two-level structure). With Definitions 2.2–2.3, DRD is *not one number but a functorial package of distinguishability*, whose components live at different structural levels and degrade differently under different forgetting operations. This removes the ambiguity of identifying "information" sometimes with a homology rank and sometimes with a spectral capacity: after the split into $\mathsf{D}^\sharp_{\mathrm{top}}$ and $\mathsf{D}^\sharp_{\mathrm{op}}$ these are not in conflict but form a hierarchy of levels.

Layer	**Object**	**Functoriality**	**What it registers**
Homological	$\widetilde{H}_k(X;\mathbb{Z}) \cong \mathbb{Z}^{\beta_k} \oplus \mathrm{Tor}_k$	covariant (f_*)	components, cycles, enclosures, torsion distinctions
Cohomological	$H^*(X;\mathbb{Z})$, $\smile$	contravariant (f^*)	intersection/ring structure, obstruction data
Homotopical	$\pi_k(X)$, Whitehead products	covariant ($\pi_k(f)$)	unstable distinctions invisible to homology
Bundle	$w_i, c_i, p_i, e, \hat{A}$ of $\mathcal{E}$	contravariant (pullback)	orientation, spin, complex/real field topology
Spectral	$\mathrm{Spec}_0, \mathrm{Spec}_{TT}, \mathrm{Spec}(D)$	invariant under (spin-)isometries	operationally resolvable capacity

2.4 Strengthened formulation: the structure fibration

The most robust way to close the object $\mathsf{D}^\sharp$ is not to force all invariants into one flat category. Homology, characteristic classes, spin data and spectra have different admissible morphisms. The correct ambient object is therefore a small category of structural packages together with its Grothendieck construction.

Definition 2.7 (Structure fibration)**.** Let **Str** be the category whose objects are packages

$$\mathfrak{s} = (\mathrm{Top}, \mathrm{CW}, \mathrm{Smooth}, \mathrm{Orient}, \mathrm{Spin/Pin}, \mathrm{Bundle}, \mathrm{Metric}, \mathrm{Operator}, \varepsilon, \Lambda, q),$$

and whose morphisms forget or weaken structure. For each $\mathfrak{s}$ let $\mathbf{Car}_{\mathfrak{s}}$ be the category of carriers preserving exactly the structure listed in $\mathfrak{s}$. The full carrier category is the Grothendieck construction

$$\mathbf{Car}^{\mathrm{fib}}_* := \int_{\mathfrak{s}\in\mathbf{Str}} \mathbf{Car}_{\mathfrak{s}}.$$

A profile is then a family of functors

$$\mathsf{D}^\sharp_{\mathfrak{s}} : \mathbf{Car}_{\mathfrak{s}} \longrightarrow \mathbf{Prof}_{\mathfrak{s}},$$

compatible with weakening of structure whenever a forgetting morphism is declared. Concretely, for a weakening $\alpha : \mathfrak{s} \to \mathfrak{t}$ the square

$$\begin{array}{ccc} \mathbf{Car}_{\mathfrak{s}} & \xrightarrow{\mathsf{D}^{\sharp}_{\mathfrak{s}}} & \mathbf{Prof}_{\mathfrak{s}} \\ {\scriptstyle U_\alpha}\downarrow & & \downarrow{\scriptstyle V_\alpha} \\ \mathbf{Car}_{\mathfrak{t}} & \xrightarrow{\mathsf{D}^{\sharp}_{\mathfrak{t}}} & \mathbf{Prof}_{\mathfrak{t}} \end{array}$$

commutes up to the declared comparison isomorphism. Here U_α forgets carrier structure and V_α forgets the corresponding profile components.

Proposition 2.8 (Layer-compatibility)**.** *Inside a fixed fibre* $\mathbf{Car}_{\mathfrak{s}}$*, isomorphisms preserve every component of* $\mathsf{D}^{\sharp}_{\mathfrak{s}}$*. Across fibres, a change of profile is not a paradox but an exhibited weakening of structure. Thus the full invariant is best written operationally as*

$$\mathsf{D}^{\sharp}_{q,\Lambda}(\mathcal{X}) = \big(\mathsf{D}^{\sharp}_{\mathrm{top},\le q}(X),\ \mathcal{C}(\mathcal{X}),\ \mathrm{Spec}_{\le\Lambda}(\mathcal{X})\big),$$

with q *a homotopy/Postnikov cutoff and* $\Lambda \sim \varepsilon^{-1}$ *a spectral cutoff.*

Remark 2.9 (Status of the closure). This closes OP1 as a mathematical construction. The formal object is the fibrewise profile $\mathsf{D}^{\sharp}_{\mathfrak{s}}$; the publishable and computable object is the truncated profile $\mathsf{D}^{\sharp}_{q,\Lambda}$; and the physical object is the same profile after causal and gravitational admissibility have been applied.

3 Carrier Categories, Forgetting Morphisms, and the Suspension Functor

3.1 Two categories of carriers

Definition 3.1 (Carrier categories)**.** $\mathbf{Car}^{\mathrm{top}}_{*}$ is the category of pointed carriers of finite CW type with basepoint-preserving continuous maps. $\mathbf{Car}^{\mathrm{op}}$ is the category of structured carriers $\mathcal{X} = (X, g, \mathfrak{s}, \mathcal{E}, \varepsilon)$ (Definition 2.3); its morphisms preserve exactly those structures needed to define the corresponding operators and spectra (isometries for $\mathrm{Spec}_0, \mathrm{Spec}_{TT}$; spin-isometries for $\mathrm{Spec}(D)$; bundle maps covering the base map for $\mathcal{C}$).

Definition 3.2 (Forgetting morphisms)**.** $\mathbf{Forg} \subset \mathrm{Mor}(\mathbf{Car})$ is the class generated by four operation types:

(i) *quotient / collapse* (identification of points or subspaces);

(ii) *deletion / restriction* (removal of points, subspaces, or components);

(iii) *change of coefficients / change of category* (e.g. passing to a homotopy-coarser category, discarding torsion, forgetting a bundle or the spin structure);

(iv) *metric–spectral coarse graining* (increasing the resolution scale $\varepsilon \mapsto \varepsilon' > \varepsilon$).

Lemma 3.3 (Coarse graining is monotone on capacity)**.** *For fixed* (X, g) *and* $\varepsilon' \ge \varepsilon$*, the band-limited spaces nest,* $\mathcal{A}_{\varepsilon'} \subseteq \mathcal{A}_{\varepsilon}$*, so every spectral capacity satisfies* $N_{\varepsilon'} \le N_{\varepsilon}$*.*

Proof. The eigenvalue condition $\sqrt{\lambda} \le 1/\varepsilon'$ implies $\sqrt{\lambda} \le 1/\varepsilon$. □

3.2 Suspension as an endofunctor

Theorem 3.4 (Transport; degree-shift conservation)**.** *The reduced suspension $\Sigma X = S^1 \wedge X$ is an endofunctor $\Sigma : \mathbf{Car}_*^{\mathrm{top}} \to \mathbf{Car}_*^{\mathrm{top}}$, and there is a natural isomorphism of functors*

$$\sigma_k : \widetilde{H}_k \implies \widetilde{H}_{k+1} \circ \Sigma, \qquad \widetilde{H}_{k+1}(\Sigma X) \cong \widetilde{H}_k(X) \quad \text{for all } k \geq 0. \tag{4}$$

Hence the reduced homological profile is shifted by one degree without loss: $\mathsf{D}_{k+1}(\Sigma X) = \mathsf{D}_k(X)$, torsion included. Suspension changes the carrier and the degree of every homological distinction; it need not preserve the full unstable homotopy profile outside the stability range specified below.

Proof. Mayer–Vietoris on the two cones covering ΣX, both contractible, with intersection homotopy equivalent to X; naturality is standard [11]. □

Theorem 3.5 (Freudenthal stabilization of homotopy data)**.** *If X is an $(n-1)$-connected CW complex, the suspension homomorphism $\pi_i(X) \to \pi_{i+1}(\Sigma X)$ is an isomorphism for $i < 2n-1$ and surjective for $i = 2n-1$ [11]. Thus suspension transports not only homological content but also homotopy-level distinguishability in the stable range, stabilizing the unstable part of $\mathsf{D}^\sharp_{\mathrm{top}}$.*

Remark 3.6 (Stability discipline)*.* Suspension is used here as a transport of reduced homology and, by Theorem 3.5, as a stable-range transport of homotopy. It is not asserted to be fully faithful on all pointed homotopy types. Thus no conclusion of the paper depends on the false statement that ΣX preserves every unstable invariant of X.

Corollary 3.7 (The fundamental class of S^3 is a suspended distinction)**.** *Along $S^0 \xrightarrow{\Sigma} S^1 \xrightarrow{\Sigma} S^2 \xrightarrow{\Sigma} S^3$,*

$$\widetilde{H}_0(S^0) \simeq \widetilde{H}_1(S^1) \simeq \widetilde{H}_2(S^2) \simeq \widetilde{H}_3(S^3) \simeq \mathbb{Z}, \tag{5}$$

carrying the distinction class of S^0 to a generator of $H_3(S^3;\mathbb{Z})$ [39]. The elementary distinction, the phase cycle, the directional enclosure, and the compact volume class are one invariant on four carriers.

Remark 3.8 (Enrichment: what higher carriers add)*.* Suspension preserves content but changes what the carrier can *do* with it. On S^1 the distinction acquires phase ($\mathrm{U}(1)$); on S^2, direction ($\mathbb{C}P^1$, the Bloch sphere); on S^3, simultaneously: compactness, simple connectivity, the non-Abelian group $\mathrm{SU}(2)$, spin structure, parallelizability, and the nontrivial Hopf bundle $S^1 \hookrightarrow S^3 \to S^2$ with $c_1 = 1$. Within the axiomatics of the companion paper, S^3 is the *minimal joint carrier of this stated package of structures* [39]—a minimality claim relative to the suspension chain and the listed conditions, not an absolute fundamentality claim.

4 Conservation and Attributed Erasure

Theorem 4.1 (Fibrewise conservation)**.** *Fix a structural package $\mathfrak{s}$ and let $f : \mathcal{X} \to \mathcal{Y}$ be an isomorphism in the fibre $\mathbf{Car}_{\mathfrak{s}}$. Then $\mathsf{D}^\sharp_{\mathfrak{s}}(f)$ is an isomorphism in $\mathbf{Prof}_{\mathfrak{s}}$. In particular:*

(i) a homotopy equivalence in the topological fibre preserves $\mathsf{D}^\sharp_{\mathrm{top}}$ but need not preserve a metric, a spectrum, a chosen field package or a chosen spin lift;

(ii) a bundle isomorphism over the base preserves the declared characteristic-class package;

(iii) an isometry preserves scalar and TT spectra with multiplicities, and a spin-isometry preserves the Dirac spectrum as well.

Thus conservation is not a global slogan: it is always conservation inside the fixed structural category whose morphisms have been declared admissible.

Proof. Homology, cohomology and homotopy data are natural in the topological fibre; characteristic classes are natural under pullback in the bundle fibre; and the relevant elliptic operators are unitarily conjugate in the metric or spin-metric fibre. The theorem is the fibrewise assembly of these standard invariance statements. Dropping any part of the structure moves to a different fibre and is therefore a forgetting operation, not a counterexample to conservation. □

Theorem 4.2 (Attributed erasure)**.** *Let $e : \mathcal{X} \to \mathcal{Y}$ be a morphism in* **Car***. If at least one component of $\mathsf{D}^\sharp(e)$ fails to be an isomorphism—a reduced homology group changes, the cup-product structure collapses, a characteristic class is lost, a stable homotopy class dies, or part of a spectrum is truncated—then e is not invertible in the corresponding structural category. Consequently every change of the profile is* attributed*: it identifies a concrete forgetting morphism in* **Forg** *(Definition 3.2), and the lost package components quantify what was erased.*

Proof. Functors preserve isomorphisms; if e had an inverse in the structural category, every induced map on the profile would be an isomorphism, contradicting the hypothesis. The generation statement is Definition 3.2.

□

Definition 4.3 (Mapping-cone loss invariant)**.** For a map $f : X \to Y$, define its cone defect by the mapping cone

$$C_f := Y \cup_f CX.$$

The degree-k homological loss and torsion loss are

$$E_k^H(f) := \operatorname{rank} \widetilde{H}_k(C_f;\mathbb{Z}), \qquad E_k^{\mathrm{Tor}}(f) := \operatorname{Tor} \widetilde{H}_k(C_f;\mathbb{Z}).$$

The exact topological erasure package is

$$\mathbb{E}(f) := \big([C_*(C_f)],\ \widetilde{H}_*(C_f),\ \tau_W(f) \text{ when the simple-homotopy torsion is defined}\big).$$

Here τ_W records simple-homotopy loss invisible to ordinary homology in acyclic cases.

Proposition 4.4 (Cone criterion for zero loss)**.** *For maps of finite CW type: if $f : X \to Y$ is a homotopy equivalence, then C_f is contractible, so contractibility of the cone is a necessary condition for zero topological erasure. Conversely, if C_f is contractible then f is an integral homology equivalence; if in addition X and Y are simply connected, f is a homotopy equivalence. The simple-connectivity hypothesis cannot be dropped: for an acyclic complex A with $\pi_1(A) \neq 0$, the map $A \to \{*\}$ has contractible cone (the suspension ΣA is simply connected and acyclic, hence contractible) without being an equivalence. If C_f is acyclic but not contractible, rank-level and torsion-level losses vanish while a simple-homotopy defect may remain.*

Proof. The first statement is standard. For the converse, the cofiber long exact sequence gives that $\widetilde{H}_*(C_f) = 0$ iff f_* is an isomorphism on integral homology; in the simply connected case Whitehead's theorem upgrades a homology equivalence to a homotopy equivalence. The counterexample follows from Hurewicz and Whitehead applied to ΣA. The final statement follows from the distinction between acyclicity and contractibility, with Whitehead torsion recording the residual simple-homotopy obstruction where applicable. □

Proposition 4.5 (Generator formulae for forgetting morphisms)**.** *For the four generator types of* **Forg** *the cone-defect package has the following concrete forms.*

(i) If $q : X \to X/A$ collapses a cofibration subspace $A \subset X$, then

$$C_q \simeq \Sigma A, \qquad \widetilde{H}_k(C_q) \cong \widetilde{H}_{k-1}(A).$$

The collapse cost is read from the shifted homology of the collapsed subcarrier.

(ii) If $j : X \setminus A \hookrightarrow X$ is deletion or restriction, then

$$\widetilde{H}_k(C_j) \cong H_k(X, X \setminus A),$$

and for good subspaces of oriented manifolds this relative group may be evaluated by Alexander or Poincare-Lefschetz duality.

(iii) If coefficients change by a ring map $R \to S$, the safe defect is computed by derived change of coefficients,

$$C_*(X; R) \longrightarrow C_*(X; R) \otimes^{\mathbf{L}}_R S,$$

and its mapping cone. This prevents torsion from being silently discarded by a non-flat coefficient change.

(iv) If spectral coarse-graining changes a cutoff Λ to $\Lambda' < \Lambda$, then for a sector P

$$E^P_{\mathrm{spec}}(\Lambda, \Lambda') = \dim \mathrm{im}(\Pi^P_{\leq \Lambda} - \Pi^P_{\leq \Lambda'}),$$

with a smoothed version obtained by replacing sharp projectors by a test function $\chi(P/\Lambda^2)$.

Consequently OP2 is not a request for one scalar loss number but for the morphism-resolved profile defect $\mathbb{E}(f)$.

Proposition 4.6 (Sector Landauer bound)**.** *If a reset operation at temperature T sends N mutually distinguishable logical sectors to one standard sector, then its average work satisfies*

$$\langle W \rangle \geq k_B T \ln N.$$

For a non-uniform prior this becomes $\langle W \rangle \geq k_B T\, \Delta H$, with ΔH the Shannon entropy decrease of the sector distribution. Higher-degree costs therefore have the safe decomposition

$$Q_T(f) \geq k_B T\, H(\textit{identified sectors}) + \Gamma_{\mathrm{geom}}(f),$$

or, for N_k equiprobable sectors,

$$Q_k \geq k_B T \ln N_k + \Gamma^{\mathrm{geom}}_k,$$

where only the entropy term is a direct Landauer consequence and Γ^{geom}_k is a protocol- and platform-dependent structural cost.

Proposition 4.7 (The minimal example)**.** *The identification $q : S^0 \to \{*\}$ kills $\widetilde{H}_0$ ($\mathbb{Z} \to 0$) and admits no homotopy inverse; deleting a point destroys the structure; and no map exists from the nonempty S^0 to $\varnothing$. The minimal distinction is erased only by an exhibited operator of type (i)–(iii), never by continuous deformation.*

Proposition 4.8 (Non-monotonicity: identifications are not pure erasure)**.** *Forgetting morphisms need not decrease total content: the contractible interval $[0, 1]$ has $\mathsf{D} \equiv 0$, yet $[0, 1]/(0 \sim 1) \simeq S^1$ has $\mathsf{D}_1 = 1$. "Erasure" and "creation" are therefore per-operation bookkeeping on the full profile, not global monotonicity statements; every change must be computed, not assumed.*

Remark 4.9 (The precise slogan). Theorems 4.1–4.2 do *not* say "nothing in the world ever disappears." They say something sharper and defensible: *within a fixed category of structure-preserving morphisms, the disappearance of an invariant is impossible; outside it, the disappearance is attributable to a specific non-invertible operator with a computable loss.*

Physical anchor 4.10 (Landauer's principle as the thermodynamic face of attributed erasure). The logically irreversible erasure of one elementary distinction (one bit) in an environment at temperature T dissipates at least $k_B T \ln 2$ [4, 6]; the bound has been approached experimentally in colloidal and nanomagnetic systems [7, 8, 9]. The topological statement (erasure requires a non-invertible operator) and the thermodynamic statement (logical irreversibility has a minimum dissipation) are two faces of one discipline: nothing erases a distinction for free. We cite this as contact with established physics; the constant $k_B T \ln 2$ is not derived from topology, and extensions to higher-degree erasures are hypotheses (Sec. 9).

5 The Operational Layer: Postulates

Postulate 1 (Carrier realization)**.** A distinction is physically realized only when represented on a carrier equipped with metric and causal structure. Without scale there is no distance; without time there is no causal relation; without causal relation there is no physically unfolded information; without distinguishability there is no observable phenomenon:

$$\text{distinction} \;\to\; \text{carrier} \;\to\; \text{order/phase} \;\to\; \text{causal relation} \;\to\; \text{reconstructible information}. \tag{6}$$

Postulate 2 (Resolution grain)**.** Within the effective model, ℓ_P (and $t_P = \ell_P/c$) is treated as the lower boundary of operationally reconstructible states: effective physical reconstruction assigns distinguishable causal-geometric states only at spatial scales $\geq \ell_P$ and temporal scales $\geq t_P$. Below the grain the model assigns no operationally verifiable state—a boundary of the reconstruction language within the effective theory, not a derived microscopic wall.

Physical anchor 5.1 (Geometric support for Postulate 2). In the companion Ricci-type flow analysis [40], a reflecting barrier at $a = \ell_P$ defines a forward-invariant region with $|\mathrm{Rm}|^2 \leq C/\ell_P^4$, mirroring a self-adjoint Wheeler–DeWitt endpoint on $a \in [\ell_P, \infty)$ with zero probability flux through $a = \ell_P$; the same work proves that natural internal forcings cannot control the singular round mode, so the grain enters as a *boundary of the effective state space*, not as a smooth regulator. Postulate 2 adopts exactly this reading and inherits its falsifier: any operationally confirmed sub-grain distinguishability refutes the postulate as stated (Sec. 14).

6 Graded Spectral Capacities on S^3_R

Counting conventions

All capacities in this section are dimension counts of band-limited eigenspaces over the field appropriate to the sector. The scalar count is the complex eigenspace dimension; for real fields it counts the corresponding real mode coordinates after the usual reality condition. The Dirac count includes both positive and negative Dirac branches and counts complex spinor components. The TT count uses the transverse-traceless symmetric tensor harmonics of the positive rough (Bochner) Laplacian $-\nabla^2$ in the convention of [14], as stated in Theorem 6.3; on the Einstein background S^3_R the Lichnerowicz operator differs from $-\nabla^2$ by a fixed curvature-proportional shift, which reindexes the cutoff M_{TT} by an $O(1)$ constant but leaves the multiplicities and the leading Weyl coefficient unchanged. These conventions are part of the definition of N_P and prevent the capacity formulae from being attacked as notation-dependent claims.

Definition 6.1 (Operational alphabet at grain ε)**.** For a closed Riemannian (spin) manifold and an elliptic operator P of Laplace or Dirac type, the band-limited space at grain ε is spanned by the eigenmodes with $\sqrt{|\lambda|} \leq 1/\varepsilon$ (Laplace type) or $|\mu| \leq 1/\varepsilon$ (Dirac type); the *mode capacity* $N_\varepsilon^{(P)}$ is its vector-space dimension: the number of independently specifiable field modes of type P resolvable at scale ε. It is not yet a bit capacity. A bit capacity is obtained only after specifying a finite operational alphabet per mode, or an energy/occupation cutoff for the associated Fock or code space.

Write $x = R/\varepsilon$ throughout.

Theorem 6.2 (Scalar capacity)**.** *On S^3_R the scalar Laplacian has eigenvalues $\lambda_n^{(0)} = n(n+2)/R^2$ with multiplicities $(n+1)^2$, $n \geq 0$ [39]. With $M_0 = \lfloor \sqrt{1+x^2} \rfloor$,*

$$N_0(\varepsilon) = \sum_{n=0}^{M_0-1} (n+1)^2 = \frac{M_0(M_0+1)(2M_0+1)}{6} \;\sim\; \frac{x^3}{3}. \tag{7}$$

Proof. $n(n+2) \le x^2 \iff (n+1)^2 \le 1+x^2 \iff n+1 \le M_0$; sum the multiplicities. □

Theorem 6.3 (Transverse-traceless tensor capacity)**.** *TT symmetric tensor harmonics on S^3_R have rough-Laplacian eigenvalues $\lambda_k^{(TT)} = \big(k(k+2)-2\big)/R^2$ with total multiplicity $2(k-1)(k+3)$ at each $k \ge 2$ [14, 15]; in the labeling $n = k+1 \ge 3$ of [14] these read $(n^2-3)/R^2$ with degeneracy $2(n^2-4)$. With $M_{TT} = \big\lfloor \sqrt{x^2+3} - 1 \big\rfloor$ (and $N_{TT} = 0$ if $M_{TT} < 2$),*

$$N_{TT}(\varepsilon) = \sum_{k=2}^{M_{TT}} 2(k-1)(k+3) = \frac{M_{TT}(M_{TT}-1)(2M_{TT}+11)}{3} \sim \frac{2x^3}{3}. \tag{8}$$

Proof. $k(k+2) - 2 \le x^2 \iff (k+1)^2 \le x^2+3 \iff k \le \sqrt{x^2+3} - 1$. The closed form follows by summing $2(k^2+2k-3)$ from $k=2$ to M_{TT} and is verified by direct evaluation at $M_{TT} = 2, 4, 9, 99$. □

Theorem 6.4 (Dirac capacity)**.** *The Dirac operator on round S^3_R has eigenvalues $\mu_n^{(D)} = \pm(n + \frac{3}{2})/R$ with multiplicity $(n+1)(n+2)$ for each sign, $n \ge 0$ [13, 39]. With $M_D = \big\lfloor x - \frac{3}{2} \big\rfloor$ (and $N_D = 0$ if $M_D < 0$),*

$$N_D(\varepsilon) = 2\sum_{n=0}^{M_D} (n+1)(n+2) = \frac{2(M_D+1)(M_D+2)(M_D+3)}{3} \sim \frac{2x^3}{3}. \tag{9}$$

Proof. $(n+\frac{3}{2})/R \le 1/\varepsilon \iff n \le x - \frac{3}{2}$; and $\sum_{n=0}^{M}(n+1)(n+2) = (M+1)(M+2)(M+3)/3$. □

Remark 6.5 (Weyl consistency and graded structure). The *leading asymptotic terms* of (7)–(9) agree exactly with the Weyl law for elliptic operators on closed 3-manifolds and on vector bundles: for the scalar, $N \sim \mathrm{Vol} \cdot \lambda^{3/2}/(6\pi^2)$ with $\mathrm{Vol}(S^3_R) = 2\pi^2 R^3$ gives $x^3/3$; the TT and Dirac coefficients are exactly twice the scalar one, matching the rank of the local field degrees of freedom (two polarizations; the two signs of the Dirac spectrum counted as spinorial sectors). The finite N's themselves come from the explicit S^3 spectra, not from the asymptotics. Two consequences deserve emphasis: capacity is naturally *graded by field type* rather than a single number, and the thresholds differ (Dirac and TT modes switch on later than scalar), which matters at moderate x.

$x = R/\varepsilon$	$N_0(x)$ scalar	$N_{TT}(x)$ TT tensor	$N_D(x)$ Dirac
1	1	0	0
2	5	0	4
3	14	10	16
5	55	76	80
10	385	696	660
50	42 925	85 456	83 300
100	338 350	675 906	666 600

Table 1: Exact graded capacities on S^3_R from (7)–(9). The scalar count is one sector of the full graded capacity; the TT and Dirac sectors activate at higher x and dominate asymptotically with coefficient 2/3.

Corollary 6.6 (Planck-grain capacity)**.** *Under Postulate 2 ($\varepsilon = \ell_P$), a quasi-local domain reconstructed as S^3_R carries the finite graded capacity $\big(N_0, N_{TT}, N_D\big)$ of Theorems 6.2–6.4 with $x = R/\ell_P$. Finiteness is a theorem given the postulates; the identification $\varepsilon = \ell_P$ is the postulate, not the theorem. Whether this kinematic count is also the* physical *count in a gravitating domain is the open problem of Sec. 11.*

7 Capacity Along Reconstruction Flow

Sharp spectral counts jump when an eigenvalue crosses the cutoff. A differentiable flow theory therefore uses the heat-trace or smooth spectral-action capacity.

Definition 7.1 (Heat-trace capacity)**.** For a smooth family of positive elliptic operators P_τ on a fixed carrier, define

$$\mathcal{N}_P(s,\tau) := \mathrm{Tr}(e^{-sP_\tau}), \qquad s = \varepsilon^2.$$

Equivalently, with a smooth cutoff χ, one may use $S_\chi(\tau) = \mathrm{Tr}\,\chi(P_\tau/\Lambda^2)$.

Proposition 7.2 (Flow derivative)**.** *Under standard smoothness and trace-class hypotheses,*

$$\partial_\tau \mathcal{N}_P(s,\tau) = -s\,\mathrm{Tr}\left((\partial_\tau P_\tau)e^{-sP_\tau}\right),$$

where the formula follows from Duhamel's identity and cyclicity of the trace, so no commutation of P_τ with $\partial_\tau P_\tau$ is assumed; and

$$\frac{dS_\chi}{d\tau} = \frac{1}{\Lambda^2}\,\mathrm{Tr}\left[\chi'(P_\tau/\Lambda^2)\partial_\tau P_\tau\right].$$

For a round $S^3_{a(\tau)}$ with $x = a/\varepsilon$, the leading sharp-count asymptotics give

$$\dot N_0 \sim \frac{a^2\dot a}{\varepsilon^3}, \qquad \dot N_{TT} \sim \frac{2a^2\dot a}{\varepsilon^3}, \qquad \dot N_D \sim \frac{2a^2\dot a}{\varepsilon^3}.$$

For the unnormalised round Ricci flow $\dot a = -2/a$, these become

$$\dot N_0 \sim -\frac{2a}{\varepsilon^3}, \qquad \dot N_{TT} \sim -\frac{4a}{\varepsilon^3}, \qquad \dot N_D \sim -\frac{4a}{\varepsilon^3}.$$

Thus reconstruction flow erases operational distinguishability sector by sector; a Planck barrier $a \geq \ell_P$ becomes a lower spectral latch rather than a verbal cutoff.

8 Causal Integration: The Physical Integration Ratio

Capacity is a property of the domain; *integrated* information is bounded by causal accessibility. On the compact causal spacetime $(S^3_R \times \mathbb{R},\ g_R - c^2dt^2)$ the causal diameter is $D_c = \pi R$ and full causal coherence is reached at

$$t_{\mathrm{coh}} = \frac{\pi R}{c} \tag{10}$$

[41].

Lemma 8.1 (Geodesic ball volume on S^3_R)**.** *The geodesic ball of radius $s \in [0, \pi R]$ has volume*

$$V_B(s) = \int_0^s 4\pi R^2 \sin^2\frac{\rho}{R}\,d\rho = 2\pi R^2 s - \pi R^3 \sin\frac{2s}{R}, \qquad V_B(\pi R) = 2\pi^2 R^3 = \mathrm{Vol}(S^3_R). \tag{11}$$

Definition 8.2 (Accessible fraction and PIR)**.** The causally accessible volume fraction at time T is

$$F(T) = \begin{cases} \dfrac{1}{\pi}\left(\dfrac{cT}{R} - \dfrac{1}{2}\sin\dfrac{2cT}{R}\right), & 0 \leq T \leq t_{\mathrm{coh}}, \\ 1, & T \geq t_{\mathrm{coh}}. \end{cases} \tag{12}$$

For an operational alphabet of capacity N_ε, the *Physical Integration Ratio* is

$$\mathrm{PIR}(T) := \frac{N_\varepsilon^{\mathrm{int}}(T)}{N_\varepsilon}, \tag{13}$$

the fraction of the capacity causally integrable by an observer by time T.

Proposition 8.3 (PIR in the uniform-support model)**.** *In the homogeneous saturation model—modes uniformly supported over the domain, integration limited only by the light cone—*$\mathrm{PIR}(T) = F(T)$*. Then:*

(a) cubic onset: $\mathrm{PIR}(T) \sim \dfrac{2\pi^2}{3}\Big(\dfrac{T}{t_{\mathrm{coh}}}\Big)^3$ *as* $T \to 0$ *(locally Euclidean growth);*

(b) exact half-time symmetry: $\mathrm{PIR}(t_{\mathrm{coh}} - T) = 1 - \mathrm{PIR}(T)$*, in particular* $\mathrm{PIR}(t_{\mathrm{coh}}/2) = \frac{1}{2}$*;*

(c) no early shortcut: *no internal observer or computation integrates the full Planck-grain capacity of Corollary 6.6 before* t_{coh}*.*

Proof. (a) Expand (12): with $u = cT/R$, $\sin 2u = 2u - \frac{4}{3}u^3 + O(u^5)$ gives $F = \frac{2}{3\pi}u^3 + O(u^5)$; substitute $u = \pi T/t_{\mathrm{coh}}$. (b) With $u' = \pi - u$: $F(u') = \frac{1}{\pi}(\pi - u + \frac{1}{2}\sin 2u) = 1 - F(u)$. (c) $F(T) < 1$ for $T < t_{\mathrm{coh}}$.

□

Proposition 8.4 (Shell-complete projector form of PIR)**.** *Let* P *be a homogeneous elliptic operator on the round* S^3_R*, and let* Π^P_Λ *be the spectral projector onto a complete set of eigenshells below the cutoff. Then for every geodesic ball* $B_s \subset S^3_R$*,*

$$\frac{\mathrm{Tr}(1_{B_s}\Pi^P_\Lambda 1_{B_s})}{\mathrm{Tr}\,\Pi^P_\Lambda} = \frac{\mathrm{Vol}(B_s)}{\mathrm{Vol}(S^3_R)}.$$

Thus, in the homogeneous shell-complete case, the volume formula for PIR *is not merely a uniform-support ansatz; it is the diagonal constancy of a symmetry-invariant spectral projector.*

Proof. The kernel of a shell-complete projector is invariant under the transitive isometry action of S^3_R. Therefore its diagonal value is constant. Integrating this constant over B_s and over all of S^3_R gives the stated ratio. □

Remark 8.5 (Inhomogeneous extension). For a general compact carrier or non-homogeneous domain the exact definition becomes

$$\mathrm{PIR}_{P,\Lambda}(B) = \frac{\int_B K^P_\Lambda(x,x)\,dV}{\int_M K^P_\Lambda(x,x)\,dV},$$

where K^P_Λ is the spectral-projector kernel. Local Weyl asymptotics then give a controlled high-cutoff approximation. This closes the homogeneous part of OP5 and turns the non-homogeneous part into a kernel-estimation problem.

Remark 8.6 (Mode-selected scalar correction). For the unit round S^3 scalar zonal harmonic of degree n,

$$\phi_n(\chi) = \frac{1}{\sqrt{2}\pi}\frac{\sin((n+1)\chi)}{\sin\chi},$$

the mass inside the geodesic ball of angular radius χ is

$$w_n(\chi) = \int_{B_\chi} |\phi_n|^2 dV = \frac{\chi}{\pi} - \frac{\sin(2(n+1)\chi)}{2\pi(n+1)}.$$

For $n = 0$ this reduces to the volume fraction $F(\chi)$; for incomplete or mode-selected alphabets the oscillatory term is the first explicit correction to shell-complete PIR.

Definition 8.7 (Reconstructible information of a quasi-local domain)**.** The *reconstructible information* of a quasi-local domain at time T is the distinguishability realized in the graded operational alphabet of Sec. 6, restricted to the causally accessible region at T (Definition 8.2), above the grain of Postulate 2. This defines the *object*; given a source distribution on the alphabet, its *quantity* is measured by the standard Shannon functional. The framework supplies the alphabet; Shannon's theory then applies unchanged.

$\tau = T/t_{\mathrm{coh}}$	$\mathrm{PIR}(\tau)$
0.00	0.000000
0.10	0.006451
0.25	0.090845
0.50	0.500000
0.75	0.909155
0.90	0.993549
1.00	1.000000

Table 2: Causal integration on S^3_R in the uniform-support model. The onset is cubic; the curve is exactly antisymmetric about the half-coherence time.

Remark 8.8 (Two limits of reconstructible information). Extraction is locked between two boundaries. *Microscopic:* the grain excludes infinite information densities; the sub-grain interval carries no weight in the reconstruction measure. *Macroscopic:* horizon formation, dilution, and expansion degrade the operational alphabet and lengthen effective coherence, limiting the causally synchronizable fraction. Within the companion reconstruction-flow reading [40], the past limit is the grain boundary of smooth geometry and the future limit is loss of recoverable quasi-local structure—*insufficient distinguishability*, not contraction.

9 Higher Erasure Costs: The Q_k Hypotheses

Landauer's bound prices the erasure of a degree-0 distinction. The graded profile of Sec. 2 invites—but does not prove—a hierarchy of prices for erasing higher-degree distinguishability. Status discipline is essential here: the following are *hypotheses defining a research programme*, not theorems.

Hypothesis 1 (Q_k family)**.** Let a physical system encode $\mathcal{N}_k$ distinguishable degree-k sectors (components, cycles, membranes/domains, or topological/fusion sectors) at temperature T. Then the minimal free-energy cost of a logically irreversible reset of the degree-k distinction obeys

$$\Delta F_k \;\geq\; k_B T \ln \mathcal{N}_k \;+\; \Gamma_k, \tag{14}$$

where $\Gamma_k \geq 0$ is the structural cost of destroying the corresponding carrier (loop tension, surface energy, topological gap, defect energy, braiding/fusion reset work). For $k = 0$, $\Gamma_0 = 0$ and $\mathcal{N}_0 = 2$ recover the Landauer bound.

Hyp.	**What is erased**	**Minimal structural form**	**Test platform**	**Status**
Q_0	binary distinction	$\Delta F_0 \geq k_B T \ln 2$	colloidal traps, nanomagnets [7, 8, 9]	established
Q_1	cycle / phase loop	$\Delta F_1 \gtrsim k_B T \ln \mathcal{N}_1 + \tau_\ell L$	superconducting vortices, persistent-current rings	hypothesis
Q_2	membrane / domain	$\Delta F_2 \gtrsim k_B T \ln \mathcal{N}_2 + \sigma A$	domain walls, interface reset	hypothesis
Q_{top}	anyon / fusion sector	$\Delta F_{\mathrm{top}} \gtrsim k_B T \ln \mathcal{D}_{\mathrm{fus}} + n\Delta_{\mathrm{gap}}$	toric-code / surface-code, FQH, Majorana platforms [16]	hypothesis

Remark 9.1 (How to test). Surface-code and toric-code architectures encode logical information in homology classes of a surface [16]; a forced reset of a nontrivial class to the trivial one is almost literally the attributed-erasure operator of Theorem 4.2 made measurable. The decisive experiments are calorimetric, not only logical: compare minimal dissipation of a fusion-sector reset with single-bit erasure at matched temperature and cycle time; for vortex and domain carriers, compare annihilation work with defect length/area. In this form the programme becomes a *topological thermodynamics of erasure* rather than a philosophy of information.

10 Quantum Information: Carrier-Resolved Resources

Normalized qubit states form $S^3 \subset \mathbb{C}^2$; ray states form $\mathbb{C}P^1 \simeq S^2$; the projection is the Hopf map, whose nontriviality ($c_1 = 1$) is the exact topological reason why global phase and Bloch direction do not factorize [39]. DRD does not replace quantum capacities; it *decomposes the carrier of quantum distinguishability*. For a d-level pure system the same carrier tower is

$$\S^{2d-1} \longrightarrow \mathbb{C}P^{d-1}, \qquad H^*(\mathbb{C}P^{d-1};\mathbb{Z}) = \mathbb{Z}[\alpha]/(\alpha^d), \qquad c_1(\mathcal{O}(-1)) = -\alpha.$$

For mixed states, the carrier is stratified by the spectrum of the density matrix; generic coadjoint orbits are flag manifolds, while degeneracies give lower-dimensional strata. Thus

$$\mathsf{D}^\sharp_Q(d) = \big(H^*(\mathbb{C}P^{d-1}), c_1(\mathcal{O}(-1)), [\omega_{FS}], \Omega_{\text{Berry}}, \text{reference-frame data}, \text{POVM alphabet}\big)$$

is the safe quantum profile: topological obstruction plus operational measurement data, not a replacement for quantum mechanics.

Definition 10.1 (Carrier-resolved qubit resources)**.** For the qubit carrier we distinguish the triple

$$C^{\text{DRD}}_{\text{qubit}} = \big(C_{\text{proj}},\, C_{\text{phase-ref}},\, C_{\text{hol}}\big)\,: \tag{15}$$

C_{proj}, distinguishability realized on the ray space S^2 (projective measurements, no phase reference); $C_{\text{phase-ref}}$, fibre-sensitive distinguishability operational only relative to an external phase reference (interferometry); C_{hol}, the global holonomy/obstruction datum of the carrier ($c_1 = 1$), a carrier-level obstruction rather than a per-state resource: it prices the impossibility of a global phase convention. In particular C_{hol} is not hidden per-state information in an absolute phase; only relative phase with respect to an operational reference frame can enter an accessible alphabet.

Remark 10.2 (Carrier-resolved reading of the Holevo bound (Interpretation))*.* The Holevo bound $I_{\text{acc}} \le \chi$ [3] is left unchanged. What DRD adds is an interpretation of *which part* of χ is accessible in a given operational setting: if the S^1 fibre is quotiented by absolute phase equivalence, only C_{proj}-type distinguishability is operational; a shared phase reference promotes part of the fibre information to the operational alphabet. This is a conceptual refinement of the alphabet, not a new theorem about channel capacity, and is labelled accordingly.

Remark 10.3 (Reference-frame discipline)*.* The phase-sensitive part of the quantum profile is operational only relative to a reference frame. For a symmetry group G, the no-reference-frame description is represented by twirling,

$$\Delta_G(\rho) = \int_G U_g \rho U_g^\dagger \, dg,$$

and the accessible ensemble distinguishability is reduced from the usual Holevo quantity to the corresponding twirled quantity. Current quantum-reference-frame work makes this relational reading explicit for general symmetries and for gauge systems, including general relativity where coordinate frames themselves are gauge-dependent [36, 37]. Crossed-product/QRF constructions further show how observer clocks and entropy definitions are linked in semiclassical gravity [38]. DRD uses these results only to specify the operational alphabet; it does not modify the Holevo theorem.

Remark 10.4 (Topological entanglement as carrier-level distinguishability (related work))*.* Topological entanglement entropy [17, 18] isolates a universal, deformation-insensitive constant in many-body entanglement—distinguishability stored at the carrier level rather than in local data, in the spirit of the bundle layer of $\mathsf{D}^\sharp_{\text{op}}$. Multi-boundary entanglement in Chern–Simons theory on link complements in S^3 [19] ties such carrier-level entanglement to link topology. We record these as adjacent results supporting the carrier viewpoint; no claim of derivation is made in either direction.

11 Volume–Area Selection and Holonomy-Compatible Boundary Admissibility

This point must not be talked away; it is made central. The graded capacities of Sec. 6 count *kinematically resolvable* band-limited modes and scale with volume, $N \sim (R/\varepsilon)^3$. Covariant entropy bounds scale with area, $S \lesssim A/4\ell_P^2$ [21]; holographic entropy formulae and their quantum extensions replace naive bulk counting by extremal or quantum-extremal generalized entropy data [27, 28]. The companion suspension paper adopts area counting for its entropy hierarchy for exactly this reason [39]. DRD does not take any boundary-projection ontology as primitive. The internal geometry of the programme remains Hopf/holonomy-based: the nontrivial phase–direction carrier is the Hopf fibration and its obstruction/holonomy data, not a claim that the bulk is ontologically projected from a boundary. The fork is therefore principled but modest: either N_ε counts all kinematic modes while gravity selects a physically realizable subspace, or the operational alphabet itself must be redefined in the presence of gravitational constraints.

Holonomy discipline

The word "boundary" in this section means an admissibility constraint or code-subspace selection rule. It is not an ontological replacement of the carrier by a boundary screen. The native DRD structure remains holonomic/Hopf-theoretic: phase, spin, c_1, Berry/Hopf holonomy and the carrier profile $\mathsf{D}^\sharp$ are primary; area bounds only constrain which parts of the kinematic alphabet can be physically realized.

Model	Idea	What counts as physical
Gravitational dressing	local modes are not gauge-invariant without gravitational anchoring	only relationally dressed observables
Boundary-code compression	kinematic alphabet restricted by boundary/area admissibility	a selected code subspace of dimension $\lesssim A/\ell_P^2$
Mode decoupling / backreaction	part of the modes does not survive self-gravity	dynamically stable modes below a collapse threshold

Definition 11.1 (Mode counts, bit capacities, and gravitational selector)**.** For a field sector P, define the raw mode count and its causally accessible part by

$$n_{\mathrm{kin}}^{(P)}(\varepsilon) := N_P(\varepsilon), \tag{16}$$

$$n_{\mathrm{causal}}^{(P)}(T) := \mathrm{PIR}_P(T)\, n_{\mathrm{kin}}^{(P)}(\varepsilon). \tag{17}$$

These are numbers of accessible modes, not bits. To convert modes into a bit capacity one must specify an operational finite alphabet per mode. Under a working normalization with $b_{\mathrm{mode}}^{(P)}$ bits per accessible mode,

$$B_{\mathrm{causal}}^{(P)}(T; b_{\mathrm{mode}}) := b_{\mathrm{mode}}^{(P)}\, n_{\mathrm{causal}}^{(P)}(T). \tag{18}$$

For a binary truncation $b_{\mathrm{mode}} = 1$; for a finite q-level alphabet $b_{\mathrm{mode}} = \log_2 q$. For an untruncated bosonic Fock space no finite bit capacity is defined by a finite mode count alone; an energy or occupation cutoff is part of the data.

The physical code-bit capacity is

$$B_{\mathrm{phys}}^{(P)}(R, \varepsilon, T) = \sup_{\mathcal{H}_{\mathrm{code}} \subseteq \mathcal{H}_{\mathrm{op}}^{(P)}(\leq \Lambda, T)} \log_2 \dim \mathcal{H}_{\mathrm{code}}, \tag{19}$$

where $\mathcal{H}_{\mathrm{op}}^{(P)}(\leq \Lambda, T)$ is the finite operational Hilbert/code space obtained after spectral cutoff, causal restriction and the chosen finite-alphabet or energy cutoff. The maximisation is constrained

by gauge invariance, gravitational dressing, generalized entropy or area bounds, and a backreaction/collapse threshold.

The gravitational selector is therefore the bit-ratio

$$\Sigma_{\rm grav}^{(P)} := \frac{B_{\rm phys}^{(P)}}{B_{\rm causal}^{(P)}} \quad (0 \le \Sigma_{\rm grav}^{(P)} \le 1), \tag{20}$$

not a ratio between a bit count and a raw vector-space dimension. A safe upper-bound form is

$$B_{\rm phys}^{(P)} \le \min\left\{ B_{\rm causal}^{(P)},\ \frac{2\pi ER}{\hbar c \ln 2},\ \frac{A}{4\ell_P^2 \ln 2} \right\}, \tag{21}$$

where the second term is the energy–radius bound [20] and the third is the area/covariant-entropy admissibility cap [21, 27, 28]. At the collapse threshold $E \sim Rc^4/(2G)$ these two external bounds have the same leading order,

$$\frac{2\pi ER}{\hbar c} \sim \frac{\pi R^2}{\ell_P^2} \sim \frac{A}{4\ell_P^2}, \tag{22}$$

so a Planck-grain volume mode count is not itself the physical bit count.

The boundary-compression ratio is correspondingly a bit-normalized quantity,

$$\mathrm{BCR}^{(P)} := \frac{A/(4\ell_P^2 \ln 2)}{B_{\rm causal}^{(P)}}. \tag{23}$$

In the one-bit-per-mode working normalization and at $\varepsilon = \ell_P$, $\mathrm{BCR}^{(P)}$ is of order ℓ_P/R for large R. Thus the volume–area tension becomes a variational selector problem rather than a verbal contradiction.

Proposition 11.2 (Bit-safe selector bound)**.** *Let $\mathcal{A}_{\rm grav}^{(P)}$ be the class of code subspaces satisfying: finite operational alphabet or energy cutoff; the gauge/diffeomorphism constraints; gravitational or relational dressing; the applicable generalized-entropy or area admissibility condition; and a backreaction/collapse threshold. Then every admissible code obeys*

$$\log_2 \dim \mathcal{H}_{\rm code} \le \min\left\{ B_{\rm causal}^{(P)}, \frac{2\pi ER}{\hbar c \ln 2}, \frac{A}{4\ell_P^2 \ln 2} \right\},$$

and therefore the selector is a contraction on bit capacity:

$$0 \le \Sigma_{\rm grav}^{(P)} \le 1.$$

Proof. The first inequality is containment of the code in the causally accessible operational Hilbert/code space. The second and third are the imposed energy–radius and area/generalized entropy admissibility constraints. Taking the supremum over $\mathcal{A}_{\rm grav}^{(P)}$ gives the stated bound and the ratio definition of $\Sigma_{\rm grav}^{(P)}$ gives the contraction property. □

Remark 11.3 (No double counting of gauge charges). In gauge theories with matter, a rigid matter symmetry, a constant asymptotic gauge transformation, a bulk current expression and a boundary flux expression can describe the same physical charge. DRD therefore counts such a charge once. It must be labelled as a bulk Noether charge, a constraint expression, a Noether identity, or a surface charge, and two expressions are identified when the field equations and boundary conditions make them equal on shell.

Remark 11.4 (Algorithmic front end for the boundary-code branch). If the boundary-code branch of $\Sigma_{\rm grav}$ is used, the relevant optimization is not a slogan about screens but an entropy-cone constrained code-subspace problem. Modern holographic-entropy methods translate linear inequalities into contraction maps and partial cubes, and recent reinforcement-learning searches use graph min-cuts to probe extreme rays of the holographic entropy cone [33, 34, 35]. In DRD this machinery supplies candidate admissibility constraints for $\mathcal{H}_{\rm code}$; it does not replace the carrier profile $\mathsf{D}^\sharp$.

Remark 11.5 (Observable-algebra reading). The dressing requirement is not optional bookkeeping. Compactly supported operators coupled to gravity require gravitational dressing, and local commuting subalgebras cease to be the right primitive notion of subsystem [29]. Recent algebraic approaches therefore study relationally dressed observables, gravitational splittings and type-II operator-algebra structures as candidates for the quantum-gravity notion of localized information [30, 31, 32]. This is precisely the mathematical habitat of Σ_{grav}.

Central remaining physical node

The strict mathematical and operational programme can be closed up to the selector $\Sigma_{\mathrm{grav}} = B_{\mathrm{phys}}/B_{\mathrm{causal}}$. The open physical question is not whether the kinematic spectral alphabet can be counted; it can. The question is which finite-alphabet or energy-cutoff code bits inside that alphabet remain gauge-invariant, holonomy-compatible, backreaction-stable, and compatible with covariant entropy bounds.

Remark 11.6 (Status). The three models are competitors, not conclusions. This is the place where the framework can fail or deepen: if all realistic realizations of capacity are gravitationally truncated to area scaling and the volume count is never physically attained, the interpretation of N_ε as physical (rather than kinematic) capacity must be revised (Sec. 14).

12 Related Definition-First Programmes and Conjectural Bridges

Information cohomology (adjacent, complementary). Baudot and Bennequin proved that Shannon entropy is, up to a multiplicative constant, the unique nontrivial 1-cocycle of an information cohomology built on categories of observables, with the chain rule as the cocycle condition; deformed coefficients yield Tsallis entropy [22]. This is a definition-first algebraicization of the information *measure*. DRD is a definition-first algebraicization of the information *alphabet and carrier*. The two programmes are complementary floors of the same tower: information cohomology derives the functional that DRD's operational alphabets are measured by. The bridge is the functor

$$\mathsf{D}^\sharp(\mathcal{X}) \longmapsto \mathbf{Obs}_{\varepsilon,T,\Sigma}(\mathcal{X}) \longmapsto H^1(\mathbf{Obs}_{\varepsilon,T,\Sigma};\mathcal{F}),$$

where objects of $\mathbf{Obs}_{\varepsilon,T,\Sigma}$ are finite spectral partitions, POVM partitions or causal/dressed subalgebras after resolution, causal and selector restrictions, and morphisms are coarse-grainings. A carrier map $f : X \to Y$ induces a pullback of observable categories whenever the structure permits it; entropy is then a cocycle on the downstream observable category, not the primitive definition of information.

Reconstruction-flow thermodynamics (conjectural bridge). Perelman's entropy functional makes Ricci flow gradient-like [23]; proposed correspondences relate that functional to Fisher information and, further, to the Bohm quantum potential [24]. If such a correspondence holds in a controlled setting, the reconstruction flow of [40] would acquire a direct statistical-distinguishability meaning, and OP3 of Sec. 15 (capacity under reconstruction flow) would gain a thermodynamic potential. We record this strictly as a *conjectural bridge*: no claim is made here that quantum mechanics is derived from geometric flow.

13 Interpretive Layer

Everything in this section is labelled *Interpretation*: a reading discipline, not physics.

Remark 13.1 (Non-emergence). The theory does not assert the absolute emergence of information, space, or geometry. S^0 gives minimal distinguishability; suspension changes its carriers (Theorem 3.4); S^3 gives the minimal joint spin/Hopf carrier of the stated package; the grain gives

the lower metric-causal boundary of physical reconstruction (Postulate 2). Within the fixed category of invertible morphisms the invariant package is neither created nor erased; across categories, every change is attributed to an exhibited operator. Cessation of a conditioned representation is not annihilation of the invariant; it is release from a conditioned carrier.

Remark 13.2 (Two protections). First, against self-being: no element of the chain—distinction, carrier, alphabet, observer—is posited as self-existing; each is given by a construction (cf. the tetralemma discipline [43]). Second, against substantializing dependence itself: having refuted self-being, one may not erect "Interdependence" as a new substance. In mathematical register: empty of own-nature, not empty of dependent functioning.

Remark 13.3 (The axis). Within the effective model, the Planck grain is the lower boundary of operationally reconstructible states; in the programme's fixed formula: *the Planck boundary is not the beginning of the universe; it is the universal lower boundary of quasi-local geometric reconstruction* [40]. In informational terms: not a first bit, but the grain below which the very notion of an operational bit is not assigned.

Core principle (Interpretation)

Information is dependently reconstructible distinguishability: a functorial package of invariants of relational structure (Definitions 2.2–2.3), preserved under structure-preserving re-description (Theorem 4.1), transported without loss across carriers (Theorem 3.4), destroyed only by exhibited non-invertible operators at a computable and thermodynamically priced cost (Theorem 4.2, Anchor 4.10), physically realized only through carrier, grain, and causal order (Postulates 1–2), graded and finite on compact domains (Theorems 6.2–6.4), and causally integrable no faster than $\mathrm{PIR}(T)$ permits (Proposition 8.3). Not self-being, not nothing: dependently reconstructible distinguishability passing through a chain of carriers.

14 Falsifiability

The mathematical layer is proved, not testable. The operational and programmatic layers have explicit failure modes.

(1) *Mathematical failure modes.* If any fibrewise functor $\mathsf{D}^{\sharp}_{\mathfrak{s}}$ fails on its declared morphism class; if the cone criterion or any formula of Proposition 4.5 fails on a standard counterexample; if the spectral conventions of Sec. 6 yield a wrong count; or if Proposition 8.4 fails for a shell-complete homogeneous sector—the strict theory does not stand. These are good risks: checkable and taste-independent.

(2) *Compact-topology searches.* The framework does not predict that the universe is S^3; it says that *if* a quasi-local domain is effectively reconstructed as a compact S^3 carrier, the graded spectral and causal consequences of Secs. 6–8 follow. Matched-circle and correlation searches constrain compact topologies [25, 26]; null results bound the radius without falsifying the mathematical core; future CMB experiments extend sensitivity.

(3) *Grain falsifier.* Any stable, operationally confirmed distinguishability *below* the effective grain boundary refutes Postulate 2 as stated, and with it Corollary 6.6.

(4) *Capacity-interpretation falsifier.* If all realistic capacity realizations are gravitationally truncated to area scaling (Sec. 11), the interpretation of N_ε as physical capacity must be revised to kinematic capacity plus a selection rule.

(5) Q_k *tests.* Calorimetric comparisons of topological-sector resets against single-bit erasure (Sec. 9) can confirm, bound, or refute each Q_k independently.

(6) *Legacy ansatz, retired.* The golden-ratio log-periodicity $\omega = 2\pi/\ln\varphi$ of the wider programme was directly tested by us against DESI DR2 BAO and Pantheon+ with null results [42]; no statement in this paper depends on it, and its current status is recorded as unsupported. A companion null-status note records the algebraic co-scaling condition $R_N = \ell_P\varphi^N$, $t_N = t_P\varphi^N$, $R_N/t_N = c$. Since $t_P = \ell_P/c$, this is a causal-admissibility identity, not a derivation of the numerical speed of light. DRD does not depend on this identity.

15 Conclusion, Publication Roadmap, and Open Problems

The paper set out to define information before using it. The result is a three-layer framework: a proved mathematical core (functorial profile, categorical conservation, attributed erasure, suspension transport with Freudenthal stabilization), an operational layer with exact graded capacities and an exact causal-integration law conditional on two explicit postulates, and a programmatic/interpretive layer strictly quarantined from both. The claimed contribution is exactly this: a definition-first, theorem-based, falsifiability-aware account of what a physically realizable alphabet of distinguishable states is, what preserves it, what destroys it and at what price, and how much of it a compact quasi-local domain can carry and causally integrate.

Publication roadmap. For refereed publication the framework separates naturally into a bundle of three papers, each with one strong thesis: (1) the mathematical core—carrier categories, $\mathsf{D}^\sharp$, conservation and attributed erasure; (2) the spectral-causal core—graded capacities N_0, N_{TT}, N_D, Weyl consistency, and PIR; (3) the physical programme—carrier-resolved quantum resources, Q_k hypotheses with experimental schemes, and the volume–area/holonomy-compatible boundary-admissibility problem. The present master document holds all three with a uniform status discipline.

Status of the open nodes. OP1 is closed at the categorical level as the fibrewise family $\mathsf{D}^\sharp_{\mathfrak{s}}$ with computable truncation $\mathsf{D}^\sharp_{q,\Lambda}$; remaining work concerns fibre-by-fibre examples, computations, and extensions; OP2 is closed at the topological level by the cone-defect package and its generator formulae, and reduced at the physical level to sector Landauer plus Γ_{geom}; OP3 is made differentiable by heat-trace capacity; OP5 is closed in the homogeneous shell-complete case by Proposition 8.4 and reduced generally to spectral-projector kernel estimates; OP6 and OP7 have precise carrier-to-observable formulations. OP4 is the unique remaining physical node: derive, compute or experimentally constrain Σ_{grav} beyond the upper-bound discipline of Proposition 11.2.

OP1. Fibrewise profile validation. Audit $\mathsf{D}^\sharp_{\mathfrak{s}}$ fibre by fibre: continuous maps, smooth maps, bundle maps, isometries and spin-isometries must never be mixed without explicitly changing fibres.

OP2. Computable cone defects. Implement $\mathbb{E}(f)$ for the four generators of **Forg**, including torsion and coefficient-change defects, and connect each nonzero component to a candidate protocol cost Γ_{geom}.

OP3. Heat-trace flow capacity. Replace sharp $N_P(\Lambda,\tau)$ by $\mathcal{N}_P(s,\tau)$ for reconstruction flows; for Berger S^3, compute the branch crossings by $SU(2)$ representation theory and compare with the round asymptotics of Proposition 7.2.

OP4. Gravitational selector. Define the bit-safe selector $\Sigma_{\mathrm{grav}} = B_{\mathrm{phys}}/B_{\mathrm{causal}}$ through gauge-invariant dressed observables, boundary-code compression subspaces, generalized entropy, the energy–radius bound and backreaction thresholds. This is the central physical problem of the programme.

OP5. Mode-resolved PIR. Extend the shell-complete theorem to incomplete scalar, TT and Dirac alphabets by explicit projector kernels, beginning with the zonal scalar formula.

OP6. Quantum carrier profile. Develop the Hopf/flag/reference-frame profile $\mathsf{D}^\sharp_Q$ for pure and mixed states and make its relation to the Holevo quantity quantitative without claiming a new capacity bound.

OP7. Information-cohomology interface. Build the functor $\mathsf{D}^\sharp(\mathcal{X}) \to \mathbf{Obs}_{\varepsilon,T}(\mathcal{X})$ and evaluate entropy cocycles only after the operational alphabet has been reconstructed.

The framework may fail; but if it fails, it fails by calculation—at a functoriality check, at a capacity formula, at a calorimetric bound, at an observational anchor—not by taste.

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